

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4643: Mobile Application Development

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1.
 - a) Security has always been a major concern for businesses, which is even greater when it comes to mobile apps. That's why Kinetic Solution, a software firm contacted you, a prominent Application Security Administrator to suggest a robust security checklist that can be followed to find common bugs and handle basic security breaches.
Recommend a generalized security checklist to them. 15
(CO4)
(PO2)
 - b) You are a new member of Google's Bug Hunting Community. Generally, you need to follow three steps to find out bugs: explore resources, report security, and at last collect your bugs as digital trophies and earn paid rewards. You have already covered resources about Android content providers, Android intent, and Android permission. Assume, a malicious application has no permission to access mobile storage and a good application returns Android intent if it can not parse it. Both of the applications use Android content providers.
Report a step-by-step approach with a diagram of a vulnerability to the Google's Bug Hunting Community. 10
(CO4)
(PO2)
2.
 - a) IUTroomBooking, an app made by some students, was launched today. You can sign in and book some rooms as a class representative or a faculty member. First, you need to provide your IUT mail and password. Then you need to confirm the credentials again and if it matches, you will be signed in. After signing in you get to see the UI. All the labels are horizontally aligned. You need to find the room number in a search bar and then click on the button to confirm the booking. After all these, everyone using the app will get to see your credentials inside your booked room.
Find two anti-patterns in the above scenario and provide their solutions. 10
(CO3)
(PO2)
 - b) ILHAM has the idea of showing every bit of information about a futsal tournament. All the information will be available after the match, like the number of goals, passes, misses, crosses, substitutes, and so on. Now he chooses to use a table-like design pattern to show all the data. The columns contain information regarding the number of goals, passes, substitutes, etc. ILHAM made a prototype and found that users need to scroll sideways (right to left) to see all the columns. The problem is that users most of the time forget which row represents which team. Propose a solution with an appropriate tabular mobile design pattern. 5
(CO4)
(PO2)
 - c) While building an app, its common to render a ScrollView but see nothing on the screen. Describe two common debugging techniques to solve this issue. 2
(CO2)
(PO1)
 - d) Discuss both the challenge and opportunity of displaying tabular data in mobile applications. 2
(CO2)
(PO1)

3. a) Replace the placeholders (#placeholderX where X can be 1 to 8) with proper keywords of react native so that the Code Snippet 1 will render the Figure 1.

```
1 import {Image, StyleSheet, View, #placeholder1} from 'react-native'
2 import PropTypes from 'prop-types'
3 import {#placeholder2} from 'react'
4
5 import AuthorRow from './AuthorRow'
6
7 const Card = ({fullname, image, linkText, onPressLinkText}) => {
8
9   const [loading, #placeholder3] = useState(#placeholder4)
10
11   const handleLoad = () => {
12     setLoading(false)
13   }
14   return (
15     <View>
16       <AuthorRow
17         fullname={fullname}
18         linkText={linkText}
19         onPressLinkText={onPressLinkText}
20       />
21       <View style={styles.image}>
22         {#placeholder5 && (
23           <ActivityIndicator
24             style={StyleSheet.absoluteFill}
25             size={'large'}
26           />
27         )}
28         <#placeholder6
29           style={StyleSheet.absoluteFill}
30           source={image}
31           onLoad={handleLoad}
32         />
33       </View>
34     </View>
35   )
36 }
37 #placeholder7 default Card
38
39 const styles = StyleSheet.#placeholder8({
40   image:{
41     aspectRatio:1,
42     backgroundColor:'rgba(0,0,0,0.02)'
43   }
44 })
```

Code Snippet 1: A JavaScript code for Question 3.a



Figure 1: An image of a custom card component Question 3.a

- b) State the differences between,
- Static and Dynamic style.
 - Stock and Jailbroken devices.

4. a) Mention the output to the console log of the Code Snippet 2.

```
1 const obj = {fname:'atiq',lname:'shafin',age:18}
2 let txt = ""
3 for(let x in obj){
4     txt += obj[x]
5 }
6 console.log(txt);
```

Code Snippet 2: A JavaScript code for Question 4.a

- b) Beginners and experts alike frequently run into the problem where a component does not render anything on the screen. State a debugging technique to solve this issue.
- c) Shafin, a newbie in software development is trying to build a touchable opacity with multiple children as element. He tries several times, only to get errors. Suggest to him a correct way to include multiple children inside a touchable opacity.
- d) Assume, a View has a child component with 1717x122 (widthxheight) dimension and the style object as its style prop, shown in the Code Snippet 4. Find the actual height and width of the View.

```
1 {
2     marginTop : 22,
3     marginBottom : 40,
4     padding : 13,
5     borderTop : 18,
6     paddingRight : 73,
7     border : 15,
8     margin : 50
9 }
```

Code Snippet 3: A JavaScript code for Question 4.d

- e) Demonstrate two approaches to include images in a mobile application with appropriate code examples.

5. a) On the Web, Multi-Step registration processes often use process bars to show users where they are and what's next. Discuss the problem faced by mobile application designers when they want to do the same and its solution with illustrations for both the problem and the solution. 15 (CO4) (PO2)

b) GiftApp, a new start-up selling gifts for different occasions like birthdays, anniversaries, etc., decided to launch their first ever mobile application. The executives want to have a visible sign-in option even when users just roaming around the application. Suggest a sign-in option for GiftApp with necessary illustrations. 5 + 5 (CO4) (PO2)

6. a) Describe two HTML events. 5 (CO1) (PO1)

b) Mention the outputs to the console log of the Code Snippet 4.

```
1 let x = 5
2 let y = '5'
3 let z = '6'
4 console.log(y<z)
5 console.log(x==y)
6 console.log(x===y)
```

6 (CO2) (PO1)

Code Snippet 4: A JavaScript code for Question 6.b

c) Suggest the seven-step process for developing any mobile application. Provide examples for each of the steps. 14 (CO1) (PO1)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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DEPARTMENT OF BUSINESS AND TECHNOLOGY MANAGEMENT

Semester Final Examination

Course No: HUM 4641

Course Title: Accounting

Summer Semester, A. Y. 2022-2023

Time : 3 hours

Full Marks : 150

Answer **all 6 (six)** questions. All questions carry equal marks. Marks of each question and corresponding CO and PO are written in the right margin within brackets.

-
- | | | | |
|----|---|----|-----------------|
| 1. | a) What are the assumptions of Cost-Volume-Profit (CVP) Analysis? Explain. | 09 | (CO1)
(PO1) |
| | b) Explain different kinds of manufacturing costs with examples. | 08 | (CO1)
(PO1) |
| | c) Distinguish between managerial and financial accounting. | 08 | (CO1)
(PO1) |
| 2. | John Corporation has collected the following information after its first year of sales. Sales were \$1,600,000 on 100,000 units, selling expenses \$250,000 (40% variable and 60% fixed), direct materials \$490,000, direct labor \$290,000, administrative expenses \$270,000 (20% variable and 80% fixed), and manufacturing overhead \$380,000 (70% variable and 30% fixed). Top management has asked you to do a CVP analysis so that it can make plans for the coming year. It has been projected that unit sales will increase by 10% next year. | 25 | (CO3)
(PO11) |
- Instructions:**
- Compute the contribution margin for the current year and the projected year, and the fixed costs for the current year. (Assume that fixed costs will remain the same in the projected year.)
 - Compute the break-even point in units and sales dollars for the current year.
 - The company has a target net income of \$200,000. What are the required sales in dollars for the company to meet its target?
 - If the company meets its target net income number, by what percentage could its sales fall before it is operating at a loss? That is, what is its margin of safety ratio?
- | | | | |
|----|---|----|-----------------|
| 3. | Meghna Industries had sales in 2016 of \$6,800,000 and gross profit of \$1,100,000. Management is considering two alternative budget plans to increase its gross profit in 2017. Plan A would increase the selling price per unit from \$8.00 to \$8.40. Sales volume would decrease by 10% from its 2016 level. Plan B would decrease the selling price per unit by \$0.50. The marketing department expects that the sales volume will increase by 100,000 units. At the end of 2016, Meghna has 40,000 units of inventory on hand. If Plan A is accepted, the 2017 ending inventory should | 25 | (CO3)
(PO11) |
|----|---|----|-----------------|

be equal to 5% of the 2017 sales. If Plan B is accepted, the ending inventory should be equal to 60,000 units. Each unit produced will cost \$1.80 in direct labor, \$1.40 in direct materials, and \$1.20 in variable overhead. The fixed overhead for 2017 should be \$1,895,000.

Instructions:

- I. Prepare a sales budget and a production budget for 2017 under each plan.
- II. Compute the production cost per unit under each plan. Why is the cost per unit different for each of the two plans? (Round to two decimals.)
- III. Compute the gross profit under each plan. Which plan should be accepted?

4. a) On April 1, Julie Spengel established Spengel's Travel Agency. The following transactions were completed during the month.

(CO2)
(PO11)

April 1: Invested \$15,000 cash to start the agency.

April 2: Paid \$600 cash for April office rent.

April 3: Purchased equipment for \$3,000 cash.

April 4: Incurred \$700 of advertising costs in the Chicago Tribune, on account.

April 8: Paid \$900 cash for office supplies.

April 9: Performed services worth \$10,000: \$3,000 cash is received from customers, and a balance of \$7,000 is billed to customers on account.

April 17: Withdrew \$600 cash for personal use.

April 19: Paid Chicago Tribune \$500 of the amount due in transaction in April (4).

April 25: Paid employees' salaries are \$2,500.

April 27: Received \$4,000 in cash from customers who have previously been billed in transaction in April (9).

Instructions: Journalize the transactions.

- b) An inexperienced bookkeeper prepared the following trial balance for Kwun 10 company. Prepare a correct trial balance, assuming all account balances are normal.

(CO2)
(PO11)

Kwun Company
Trial Balance
December 31, 2018

Accounts Title	Debit	Credit
Cash	\$14,800	
Prepaid Insurance		\$3,500
Accounts Payable		3,000
Unearned Revenue	2,200	
P. Kwun, Capital		13,000
P. Kwun, Drawing		4,500
Service Revenue		25,600
Salaries Expense	18,600	
Rent Expense		2,400
	<u>\$35,600</u>	<u>\$52,000</u>

5. The adjusted trial balance columns of the worksheet for Thao Company, owned by D. Thao, are as follows. 25 (CO2) (PO11)

THAO COMPANY
Worksheet
For the Year Ended December 31, 2017

Account Titles	Adjusted Trial Balance	
	Debit (\$)	Credit (\$)
Cash	5,300	
Accounts Receivable	10,800	
Supplies	1,500	
Prepaid Insurance	2,000	
Equipment	27,000	
Accumulated Depreciation—Equipment		5,600
Notes Payable		15,000
Accounts Payable		6,100
Salaries and Wages Payable		2,400
Interest Payable		600
Owner's Capital		13,000
Owner's Drawings	7,000	
Service Revenue		61,000
Advertising Expense	8,400	
Supplies Expense	4,000	
Depreciation Expense	5,600	
Insurance Expense	3,500	
Salaries and Wages Expense	28,000	
Salaries and Wages Expense Interest Expense	600	
Totals	<u>\$103,700</u>	<u>\$103,700</u>

Instructions:

- I. Prepare an income statement.
- II. Prepare a balance sheet. (Note: \$5,000 of the notes payable become due in 2018.) Thao did not make any additional investments in the business during the year.
- III. Prepare the closing entries. Use J14 for the journal page.
- IV. Prepare a post-closing trial balance.

6. a) Kingston Company uses job order costing and applies overhead to production based on direct labor costs. On January 1, 2017, Job 50 was the only job in process. The costs incurred prior to January 1 on this job were as follows: direct materials \$20,000, direct labor \$12,000, and manufacturing overhead \$16,000. As of January 1, Job 49 had been completed at a cost of \$90,000 and was part of finished goods inventory. There was a \$15,000 balance in the Raw Materials Inventory account. During the month of January, Lott Company began production on Jobs 51 and 52, and completed Jobs 50 and 51. Jobs 49 and 50 were also sold on account during the month for \$122,000 and \$158,000, respectively. The following additional events occurred during the month.

25 (CO3)
(PO11)

I. Purchased additional raw materials of \$90,000 on account. Incurred factory labor costs of \$70,000. Of this amount \$16,000 related to employer payroll taxes.

II. Incurred manufacturing overhead costs are as follows: indirect materials \$17,000, indirect labor \$20,000, depreciation expense on equipment \$12,000, and various other manufacturing overhead costs on account \$16,000.

III. Assigned direct materials and direct labor to jobs as follows.

Job No	Direct materials	Direct labor
50	\$10,000	\$5,000
51	39,000	25,000
52	30,000	20,000

Instructions:

- I. Calculate the predetermined overhead rate for 2017, assuming Lott Company estimates total manufacturing overhead costs of \$840,000, direct labor costs of \$700,000, and direct labor hours of 20,000 for the year.
- II. Open job cost sheets for Jobs 50, 51, and 52. Enter the January 1 balances on the job cost sheet for Job 50.
- III. Prepare the journal entries to record the purchase of raw materials, the factory labor costs incurred, and the manufacturing overhead costs incurred during the month of January.
- IV. Prepare the journal entries to record the assignment of direct materials, direct labor, and manufacturing overhead costs to production. In assigning manufacturing overhead costs, use the overhead rate calculated in (1). Post all costs to the job cost sheets as necessary.
- V. Total the job cost sheets for any job(s) completed during the month. Prepare the journal entry (or entries) to record the completion of any job(s) during the month. Prepare the journal entry (or entries) to record the sale of any job(s) during the month.
- VI. What is the balance in the Finished Goods Inventory account at the end of the month? What does this balance consist of?
- VII. What is the amount of over- or under-applied overhead?

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SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 100

CSE 4615: Wireless Networks

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

-
- | | | |
|----|--|-------------------------|
| 1. | a) Discuss the relationship between different inter-frame spaces used in IEEE 802.11. Describe the use of different inter-frame space. | 4 |
| | b) How does flow control differ from congestion control? What are the changes required for both flow control and congestion control when they are applied in a wireless network? | 4 |
| | c) Skype offers a service that allows you to make a phone call from a PC to an ordinary phone. This means that the voice call must pass through both the Internet and a telephone network. Discuss how this might be done. | 4 |
| | d) Wireless network follows a layered architecture model for data communication. The main difference between wired and wireless technology is how the signal is passed through the transmission medium which is handled by physical layer. For wireless communications, is it mandatory to change other layers to transmit data properly? Briefly discuss on it. | 4 |
| 2. | a) Explain the reason for poor performance of the TCP protocol in wireless networks. How the performance of TCP protocol in wireless network can be improved? | 4 |
| | b) What are the challenges of DTN compared to TCP/IP? Mention some applications of DTN and explain how it is required for those application. | 4 |
| | c) In wireless network, the collision cannot be identified by measuring the signal power at the time of transmission. CSMA/CA avoid collision in wireless network. How can CSMA/CA protocol avoid collision if the collision cannot be detected? | 4 |
| | d) Explain the hidden terminal problem and exposed terminal problem with necessary diagrams. How can you resolve these problems? | 4 |
| 3. | a) What are the metrics to indicate lifetime of WSN? How the LEACH protocol enhance network lifetime? With a labelled diagram explain how LEACH protocol works. | 10
(CO1)
(PO1) |
| | b) Explain the working procedure of S-MAC protocol for WSN. With necessary diagram explain the Synchronized Island problem. | 10
(CO1)
(PO1) |
| 4. | a) Explain the basic operation of the Ad-hoc On-Demand Distance Vector (AODV) routing protocol. Describe the role of sequence numbers in AODV routing. | 7 + 3
(CO1)
(PO1) |
| | b) Explain the basic operation of the Dynamic Source Routing (DSR) protocol in Mobile Ad Hoc Networks (MANETs). How does it differ from AODV protocol? | 6 + 2
(CO1)
(PO1) |

5.

a) How Energy Harvesting Wireless Sensor Networks (EH-WSN) is different from the traditional Wireless Sensor Network (WSN)? Are there any changes required in EH-WSN compared to traditional WSN? If so, briefly discuss on it.

5
(CO1)
(PO1)
- b) Figure 1 represents a Energy Harvesting Wireless Sensor Networks (EH-WSN). The network parameters of the nodes and the links are listed in Table 1 and Table 2 respectively. Calculate the route from A to E following the wastage aware routing protocols during morning and noon time. [Hints: You need to show the calculation in each node].

10
(CO2)
(PO2)

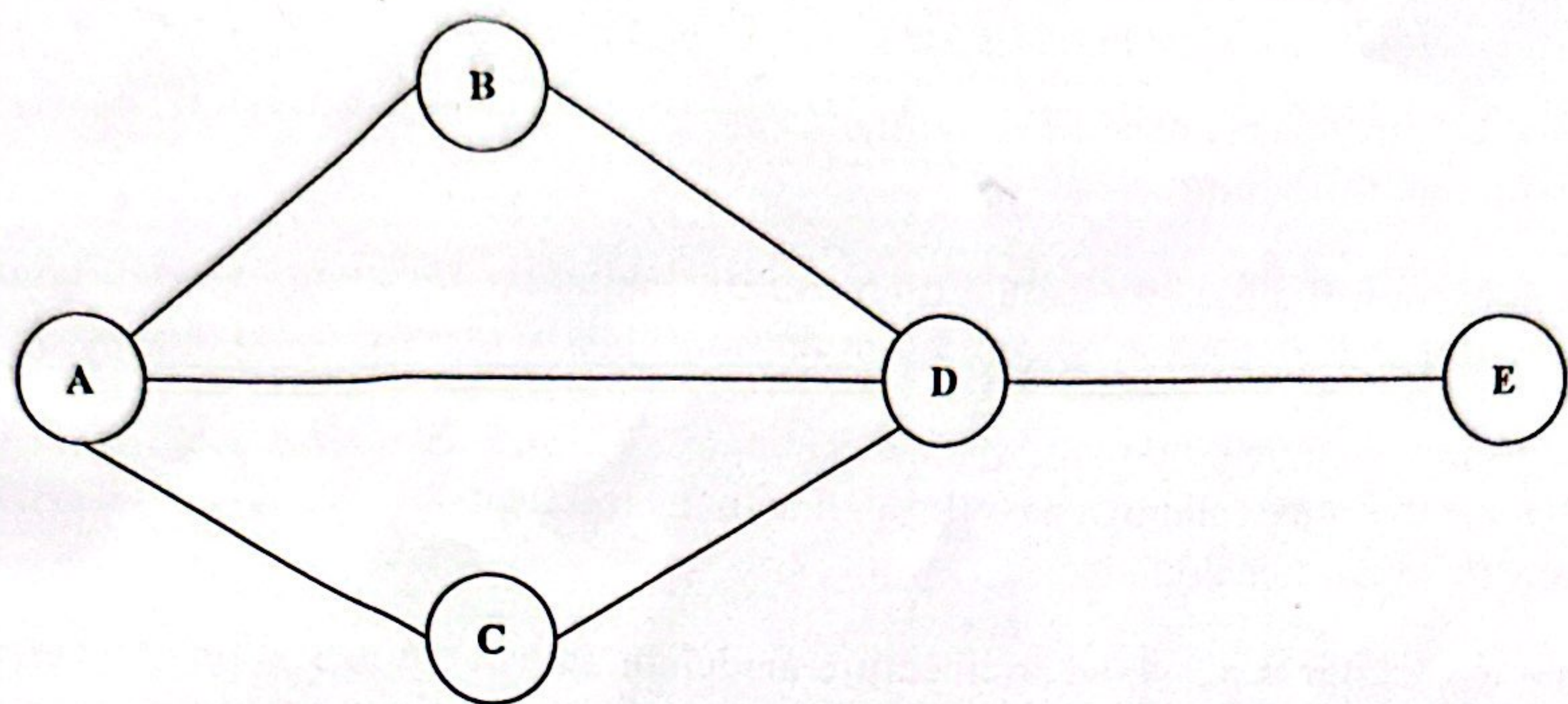


Figure 1: Topology for Energy Harvesting Wireless Sensor Networks for Question 5.b

Table 1: Network Parameters of Nodes for Question 5.b

Node	Current Battery Level	Expected Energy Harvest in ΔT , E_h (Morning Time)	Expected Energy Harvest in ΔT , E_h (Noon Time)
A	0.90 B	0.15 B	0.10 B
B	0.95 B	0.25 B	0.20 B
C	0.80 B	0.20 B	0.30 B
D	0.90 B	0.20 B	0.15 B
E	0.79 B	0.07 B	0.07 B

Table 2: Network Parameters of Nodes for Question 5.b

Link	Expected Energy Consumption in ΔT , E_h (All time)
A-B	0.15 B
A-C	0.20 B
B-D	0.25 B
C-D	0.10 B
D-E	0.15 B
A-D	0.15 B

6.

a) Ananda is designing a routing protocol for a wireless network. In calculating the best route, he used standard hop count. Is there any problem in the design choice of Ananda? Describe the reasoning behind your answer. What could be other alternatives? Explain why these alternatives are better?

5
(CO3)
(PO2)

b) Figure 2 represents the topology for a Wireless Metropolitan Area Network (WMAN). The measured forward and reverse delivery ratio is listed in Table 3. Using Expected Transmission Count (ETX) as the routing metric, calculate the route from each nodes to other nodes of the network and mention the route ETX. 10
(CO2)
(PO2)

Table 3: The measured forward and reverse delivery ratio of each link for Question 6.b

Link	Forward Delivery Ration R_{fwd}	Reverse Delivery Ration R_{rev}
A-B	0.85	0.92
A-C	0.70	0.93
A-D	0.92	0.87
B-D	0.76	0.73
C-D	0.88	0.90
C-F	0.77	0.87
D-E	0.88	0.66
E-F	0.79	0.92

$S = 1$
 $B = 1$

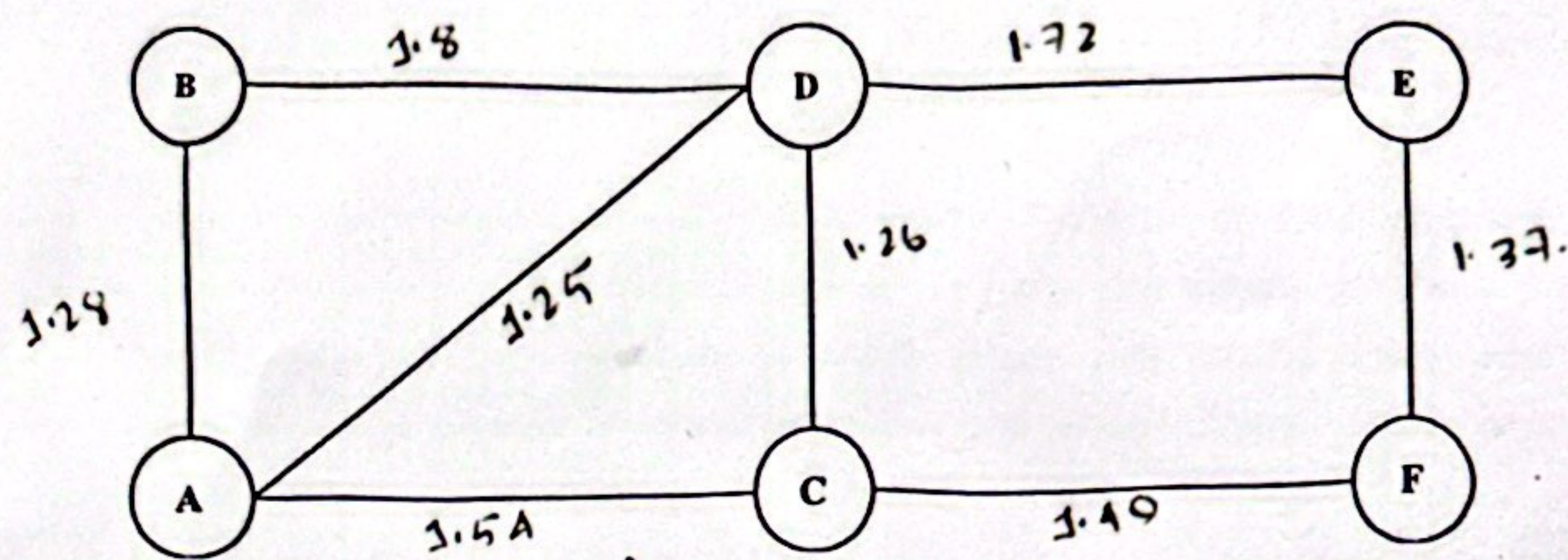


Figure 2: A Topology denoting a Wireless Metropolitan Area Network (WMAN) for Question 6.b

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SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4619: Peripherals and Interfacing

Programmable calculators are not allowed. Do not write anything on the question paper.
Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What is an Embedded System? How does it differ from typical Computer Systems? 3 + 7
(CO1)
(PO1)
 - b) Draw the block-diagram of an embedded system considering a designer's view. Also, write down the customer functions for an embedded system. 4 + 4
(CO2)
(PO2)
 - c) Draw an optimized block diagram of a PCB considering a set of peripherals with two-set 4-LEDs, two-set 7-Segment Displays and one-set 8 × 8 Dot-Matrix Display connected to a single CPU using multiple IC 8255 PPIs. 7
(CO4)
(PO3)
2. a) Explain the importance of a priority resolver IC. Differentiate between Daisy-Chain and Multi-level Bus architectures with proper explanation. 3 + 7
(CO2)
(PO2)
 - b) Make the lists of the operating modes and operation command words of 8259 PIC. Using a block diagram, explain the functions of different registers of 8259 PIC. 4 + 4
(CO2)
(PO1)
 - c) Derive all data bits for Port-A, Port-B and Port-C of IC 8255 considering the (4, 5) and (5, 4) LEDs selected with green colors in an 8 × 8 Dot-Matrix Display. You can assume any kind of logic bit (i.e., 0 or 1) for unique selections for port data. 7
(CO4)
(PO2)
3. a) Explain why Programmed I/O is not suitable for Disk RD/WR. Listing different applications for Disk RD/WR, explain a suitable solution in this regard. 5 + 5
(CO2)
(PO1)
 - b) Using an appropriate block diagram, mention how the IC 8237 DMA Controller uses the system buses. 8
(CO2)
(PO1)
 - c) Define UART and USART. Draw the interfacing structure of IC 8251. 4 + 3
(CO2)
(PO1)
4. a) Explain Wired-AND concept. CAN bus protocol removes a complete graphs connection complexity for an embedded system Justify the statement with proper explanation. 3 + 7
(CO3)
(PO1)
 - b) Write short notes on following frames type of CAN bus: 4 + 4
(CO3)
(PO1)
 - i. Remote frame
 - ii. Error frame

- c) Suppose, in a CAN bus protocol, four bytes of data (10101001, 11110000, 00111011, and 00010001) from one IC are consecutively transmitted to another IC. Considering the number of bits, $n = 8$ for Check-sum approach, show the operations for error-free data transfer between the ICs. 7
(CO3)
(PO3)
5. a) Explain by Arbitration in BUS protocols. "Multi-master concept exists for I²C bus" — verify the statement with proper explanation. 3 + 7
(CO3)
(PO1)
- b) Explain different types of acknowledgement and negative-acknowledgement transmissions in I²C bus with appropriate signal diagrams. 8
(CO3)
(PO1)
- c) Draw the data formats of I²C protocol when the Master IC reads and writes to/from Slave IC. 7
(CO3)
(PO3)
6. a) Define Jitter? Explain the concepts of Isochronous interfacing with appropriate examples. 3 + 7
(CO1)
(PO1)
- b) Write short notes on the following wireless interfacing concepts with their comparison table: 5 × 3
(CO3)
(PO1)
- IrDA
 - Bluetooth
 - IEEE 802.11
 - IEEE 802.15.4
 - LoRaWAN

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
FULL MARKS: 150

CSE 4631: Digital Signal Processing

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) A low-pass FIR filter is to be designed with the following specifications: passband 10 kHz, stopband 11 kHz, stopband attenuation 50 dB, and a sampling frequency of 44 kHz. Assume that having better stopband attenuation is more important than having faster rolloff. Determine the impulse response $h(n)$ for this filter. 15
(CO2)
(PO1)
 - b) Compute the convolution of the following input signal $x(n)$ with the impulse response $h(n)$. 10
(CO2)
(PO1)

$$x(n) = u(n+1) - u(n-4) - \delta(n-5)$$

$$h(n) = [u(n+2) - u(n-3)] \cdot (3 - |n|)$$
2. a) The frequency response $H(n)$ of a system is given in the polar form as follows: 9 + 4
(CO3)
(PO2)

$$\text{Mag}H = \{80, 31.305, 8\}$$

$$\text{Phase}H = \{0, -0.4636, 3.1416\}$$

where $\text{Phase}H(n)$ is given in radians.
Find the impulse response and step response of this system.

 - b) Explain how time domain aliasing and frequency domain aliasing occur within a signal. 6
(CO3)
(PO2)
 - c) Suppose you are given a 60 point time domain audio signal and you wish to generate a higher resolution version of this signal having 1000 points. Describe the frequency domain interpolation procedure needed to obtain your desired result. 6
(CO3)
(PO2)
3. a) Describe the problems associated with using the polar notation to represent the frequency domain of a signal. 7
(CO1)
(PO1)
 - b) Identify the factors that limit the frequency resolution of a signal. 8
(CO1)
(PO1)
 - c) The frequency domain representation of the input, $X(n)$, and the output, $Y(n)$, of a system is given in the rectangular form as follows: 10
(CO2)
(PO1)

$$X = \{80 + 0i, 28 - 14i, -8 + 0i\}$$

$$Y = \{-80 + 0i, -70 + 140i, 24 + 0i\}$$

Find the frequency response of the system in rectangular form.
4. a) Illustrate the effect on the phase of a signal due to a right shift or left shift in the time domain with relevant examples. 8
(CO1)
(PO1)

- b) Show how the bit reversal sorting algorithm is used to find the time domain decomposition of a 16 point signal in the Fast Fourier Transform (FFT) algorithm. 7
(CO2)
(PO1)
- c) Explain how the FFT butterfly functions as a fundamental computation unit in the FFT synthesis step. Illustrate how the FFT butterfly is used to combine two 8 point frequency spectra into a single 16 point frequency spectra. 5 + 5
(CO2)
(PO1)
5. a) Explain how FFT is more advantageous over DFT in terms of both time complexity and accuracy. 9
(CO1)
(PO1)
- b) Suppose the filter kernel of a low-pass filter having the cutoff frequency 200 Hz is given as $h_l = \{-1, 0, 1, 2, 1, 0, -1\}$. Assuming that the sampling frequency is 1 kHz, Calculate the filter kernel for the following filters:
- i. A high-pass filter having a cutoff frequency at 300 Hz. 5
(CO2)
(PO1)
- ii. A band-pass filter that allows frequencies in the range of 200 Hz to 300 Hz. 6
(CO2)
(PO1)
- iii. A band-reject filter that stops frequencies in the range of 200 Hz to 300 Hz. 5
(CO2)
(PO1)
6. a) Suppose you are given an arbitrary time domain input signal, $x(n)$, and your DFT function is providing the following frequency domain representation after calculating a 16 point real DFT of the input signal:
 $X(n) = \{4 + 2i, 5 - 2i, 6 + 6i, 7 - 7i, 8 + i, 7 + i, 6 - 2i, 5 + 2i\}$
Justify how you know that your implementation of the DFT function is not correct. 8
(CO3)
(PO2)
- b) The following input-output pairs have been observed during the operation of a linear system:
- $$x_1(n) = \{-1, 2, 1\} \xrightarrow{\tau} y_1(n) = \{1, 2, -1, 0, 1\}$$
- $$x_2(n) = \{1, -1, -1\} \xrightarrow{\tau} y_2(n) = \{-1, 1, 0, 2\}$$
- $$x_3(n) = \{0, 1, 1\} \xrightarrow{\tau} y_3(n) = \{1, 2, 1\}$$
- Here, the leftmost point is the origin for each of the signals and τ represents the system.
Provide justifications regarding the time invariance of the system.
- c) Explain why amplitude modulation is used in DSP. Describe its effect on the corresponding frequency domain of the modulated signals. 4 + 6
(CO3)
(PO2)

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
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Department of Computer Science and Engineering (CSE)

SEMESTER FINAL EXAMINATION
 DURATION: 3 HOURS

SUMMER SEMESTER, 2022-2023
 FULL MARKS: 150

CSE 4621: Machine Learning

Programmable calculators are not allowed. Do not write anything on the question paper.

Answer all 6 (six) questions. Figures in the right margin indicate full marks of questions with corresponding COs and POs in parentheses.

1. a) What are the benefits of using the convolution operation in Convolutional Neural Networks (ConvNets) over regular Neural Networks (NNs)? 6
(CO1)
(PO1)
 - b) If we initiate all weights in a Neural Network to a constant c , we suffer from the weight symmetry problem after each update. Again, following the standard approach of random initialization with $W_{ij} \simeq N(0, \sigma^2)$ will cause the output to converge to zero or diverge depending on whether the σ value is too small or too large. In this scenario, how can you initialize the weights? 4
(CO1)
(PO1)
 - c) Consider the following hypothetical ConvNet architecture defined by series of different layers (i.e. Convolutional layer, RELU function, Max Pooling layer, Flattening, FC layer, Softmax layer). Calculate the shape of the output volume and the number of parameters (weights and bias) at each layer. 12
(CO1)
(PO1)

You should write the output shapes in the format (H, W, C) , where H , W , C are the height, width and channel dimensions, respectively. Unless specified, assume valid convolution where appropriate.

Starting with an input sized $64 \times 64 \times 3$:

CONV3-16 > RELU > CONV3-32 > RELU > POOL2 > CONV5-16 > CONV5-32 > POOL-2 > Flatten > FC28 > FC14 > Softmax-5

Here the notations follow the convention given below:

 - i. CONV f - N denotes a convolutional layer with N filters, each them of size $f \times f$.
 - ii. POOL- n denotes a $n \times n$ max-pooling layer with stride of n and 0 padding.
 - iii. FLATTEN denotes to flattenning the feature vector.
 - iv. FC- N denotes a fully-connected layer with N neurons.
 - v. Softmax- N denotes Softmax function with N neurons.
 - d) Why are the residual connections often used in deep neural networks? 3
(CO1)
(PO1)
2. a) During decision tree generation for classification, instead of taking a binary split for the numeric attribute, we use ternary split using two thresholds w_{ma} and w_{mb} . In other words, we use three potential branches where samples can take j_{th} branch according to the following conditions:

$$x_k < w_{ma}; \quad w_{ma} \leq x_k \leq w_{mb}; \quad x > w_{mb}$$

Propose a modification of the tree induction method along with impurity measure to learn those two thresholds.

 - b) Compare between Discriminative model and Discriminant function with examples. 7
(CO2)
(PO2)

- c) From the training data in Table 1 classify the tuple $x = [youth, low, no, fair]^T$ using Naïve Bayes classifier. Calculate individual class-conditional probability for each class. 10 (CO1) (PO1)
3. a) What properties does an impurity test function need to satisfy? Give an example of impurity test function and draw its response graph for a two-class problem. 5 (CO1) (PO1)
- b) Consider the Table 1 which presents a training set of class-labeled tuples from the **AllElectronics** customer database. In this example, each attribute is discrete valued. The class label attribute, *buys_computer* has two distinct values namely, {yes, no}. Therefore, there are two distinct classes. Let class C_1 correspond to *yes* and class C_2 correspond to *no*. There are nine tuples of class *yes* and five tuples of class *no*. RID represents the record/tuple number, which is not useful for our task. Generate a decision tree from this database along with all calculations for split generation. 20 (CO1) (PO1)

Table 1: Training samples from the AllElectronics customer database for Question 3.b

RID	age	income	student	credit_rating	Class:buys_computer
1	youth	high	no	fair	no
2	youth	high	no	excellent	no
3	middle_aged	high	no	fair	yes
4	senior	medium	no	fair	yes
5	senior	low	yes	fair	yes
6	senior	low	yes	excellent	no
7	middle_aged	low	yes	excellent	yes
8	youth	medium	no	fair	no
9	youth	low	yes	fair	yes
10	senior	medium	yes	fair	yes
11	youth	medium	yes	excellent	yes
12	middle_aged	medium	no	excellent	yes
13	middle_aged	high	yes	fair	yes
14	senior	medium	no	excellent	no

4. a) Imagine you are working on a project to analyze customer purchasing behavior for a large retail chain with millions of transactions recorded over several years. The dataset contains a wide range of features such as customer demographics, purchase history, time of purchase, product categories, and more. However, the dataset is extremely high-dimensional, with thousands of features, making it challenging to extract meaningful insights and build predictive models efficiently. As a machine learning expert, you have been tasked with developing a customer categorization model to identify distinct customer groups based on their purchasing patterns. However, due to the high dimensionality of the dataset, traditional machine learning algorithms like clustering or classification models struggle to effectively process and extract insights from the data. How can you leverage Principal Component Analysis (PCA) to address the challenges posed by the high-dimensional nature of the dataset? 8 (CO1) (PO1)
- b) What are the implications of having a *positive*, *negative*, or *zero* covariance between variables in terms of their linear relationship and impact on statistical analysis? 3 × 3 (CO2) (PO2)
- c) Let an orthonormal transformation $y = \Phi^T x$, where the matrix Φ contains all eigenvectors. Show that for orthonormal transformations, Euclidean distances are preserved, i.e., $\|y\|^2 = \|x\|^2$. 8 (CO1) (PO1)

5. a) Explain why k-means clustering algorithm may not find the best solution? How can you choose the right value of k? 5 + 3
(CO1)
(PO1)
- b) Consider the data set as given in Table 2 consisting of the scores of two variables on each of six individuals. 6 + 6
(CO1)
(PO1)

Table 2: Dataset for Question 5.b

Sample	x_1	x_2
1	1.0	1.0
2	1.5	2.0
3	3.0	4.0
4	7.0	7.0
5	3.5	5.0
6	4.5	5.0

Apply the *k-medoids* clustering algorithm with the value $k = 2$. Provide all required calculations up to two cluster-center updates.

- c) "Both *k-means* and *k-medoids* clustering algorithms produce convex shaped clusters" — Do you agree or disagree? Justify your answer. 5
(CO2)
(PO2)
6. a) The Support Vector Machine (SVM) is a highly accurate classification method. However, SVM classifiers suffer from slow processing when training with a large set of data tuples. Discuss how to overcome this difficulty and develop a scalable SVM algorithm for efficient SVM classification in large data sets. 7
(CO1)
(PO1)
- b) Consider a Support Vector Machine to classifier the following training data for a two-class problem given in Table 3:

Table 3: Training data for Question 6.b

Class	x_1	x_2
+	1	1
+	2	2
+	2	0
-	0	0
-	1	0
-	0	1

- i. After plotting these six training points (use a separate graph paper), construct the weight vector for the optimal hyperplane, and calculate the optimal margin width. 7
(CO1)
(PO1)

- ii. If you remove one of the support vectors does the size of the optimal margin decrease, or stay the same, or increase? Mention explicitly which one you have removed. 3
(CO2)
(PO2)

[Note: You do not need to calculate the solutions by solving, rather find the answers by inspecting the graph.]

- c) How does the kernel trick work in SVM classifier? Give an example with the polynomial kernel that can be effectively utilized to handle non-linearly separable data sets. 4 + 4
(CO1)
(PO1)